

Electronic Supplement: A Systematic Survey and Taxonomy of Prompt Engineering Evaluation Frameworks for Large Language Models

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About this supplement. This file contains Appendices A, B, and C of the main paper. Cross-references to sections and tables in the main paper are given as static text (e.g., “Section 7.5 of the main paper”) because this supplement compiles independently.

Appendices

A Evaluation Design Checklist for Prompt Engineering Studies

The following checklist is intended for use before finalizing an evaluation protocol for a prompt engineering study. Each group corresponds to one taxonomy dimension. Items are written as yes/no questions; a “no” answer does not disqualify a study but should be accompanied by explicit justification in the paper.

D1 - Evaluation Method

- Have you explicitly named the evaluation method your study uses (benchmark, human evaluation, LLM-as-judge, ablation study, or user study) in the methodology section?
- If you use benchmark evaluation, have you verified that the chosen benchmark’s task type and answer format match your prompting technique’s target capability?
- If you use LLM-as-judge, have you reported the judge model, version, prompt template, and scoring rubric in sufficient detail for replication?
- If you use human evaluation, have you reported annotator qualifications, inter-annotator agreement, and the number of evaluated items?

D2 - Task Scope

- Have you stated the task domain (e.g., reasoning, code generation, clinical QA, multilingual generation) explicitly, and confirmed it is reflected in your evaluation setup?
- If your technique targets a specialized domain (clinical, legal, multilingual), have you used or adapted evaluation instruments designed for that domain rather than borrowing general NLP metrics without justification?
- If you report results on multiple task types, have you analyzed performance separately per task type rather than reporting only an aggregate?

D3 - Metric Type

- Have you stated which metric type your study uses (reference-based, model-based, LLM-as-judge, or task-specific) and explained why it is appropriate for the task?

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- If you use reference-based metrics (BLEU, ROUGE, exact match), have you confirmed that a gold reference answer or label exists for every evaluated instance?
- If your task involves open-ended generation without a single correct answer, have you justified why your chosen metric adequately captures the relevant quality dimensions?
- Have you reported metric values with confidence intervals or significance tests where sample size permits?

D4 - Automation Level

- Have you characterized the automation level of your evaluation (fully manual, human-in-loop, hybrid, or fully automated)?
- If your evaluation is fully automated and uses a proprietary LLM as judge or scorer, have you documented the API version or snapshot date so that the exact judge state is recoverable?
- If any part of the evaluation involved human review (e.g., spot-checking automated scores), have you described the sampling procedure and what was reviewed?
- Have you considered whether a change in the judge model between submission and peer review could alter reported results?

D5 - Evaluation Scope

- Have you stated the evaluation scope explicitly (single-model/dataset, cross-model, cross-dataset, cross-domain, or cross-lingual)?
- If you report a technique as generally effective, have you evaluated it on at least two distinct models or two distinct datasets to support that generalizability claim?
- If results are reported on a single model, have you framed conclusions accordingly (e.g., “on GPT-4” rather than “across LLMs”)?
- If your study targets multilingual or cross-lingual evaluation, have you confirmed that your metric and judge model perform reliably in the target languages?

D6 - Robustness and Prompt Sensitivity

- Have you reported performance under at least one semantically equivalent paraphrase of the headline prompt, or explained why such reporting is not applicable?
- If your evaluation pipeline includes an LLM-as-judge component, have you reported the judge’s sensitivity to scoring-prompt rephrasing (e.g. JSS or equivalent) [8, 39]?
- For headline numerical claims (“technique X improves accuracy by Y%”), have you confirmed that the claimed improvement survives the paraphrase variance, or framed the claim accordingly (“on this prompt phrasing”)?

D7 - Reproducibility and Contamination Resistance

- Have you released the prompts, exemplar sets, model configurations, and random seeds needed for a third party to re-run your evaluation?
- If any LLM-as-judge component is involved, have you reported the judge model identity, API version or snapshot date, and judge prompt template?
- Have you verified, by means stronger than a generic claim, that the test set used did not appear in any training corpus the underlying model has plausibly seen (e.g., dynamic benchmark construction [65], contamination probes, or held-out construction)?
- If a contamination check is not feasible, have you reported the limitation explicitly?

This checklist is available in machine-readable form at <https://github.com/rohithreddybc/PromptEvalTaxonomy>. Researchers may copy and adapt it for their own study preregistration or supplementary materials, with citation to this work.

B Representative Per-Paper Taxonomy Mapping (All Seven Dimensions)

Table B.1 presents a representative cross-section of the 151-paper corpus coded across all seven taxonomy dimensions. Ten papers are shown here – one per major paper category, selected to illustrate the range of configurations in the corpus. The complete 151-paper machine-readable dataset is included as a review-time snapshot with the submission; a citable tagged version will be released at acceptance to <https://github.com/rohithreddybc/PromptEvalTaxonomy> (see the Data Availability statement in the main paper). D6 (Robustness) and D7 (Reproducibility) codings were dual-coded with inter-rater $\kappa = 0.76$ and 0.78 respectively (see Section 7.5 of the main paper). Abbreviations: BM = Benchmark; LJ = LLM-as-judge; HE = Human eval; Abl = Ablation; TS = Task-specific; MB = Model-based; Hyb = Hybrid; Auto = Automated; HiL = Human-in-loop; CM = Cross-model; CD = Cross-domain; CDs = Cross-dataset; SM = Single-model; Sng = Single-prompt; MAg = Multi-prompt aggregate; SB = Sensitivity-bounded; AR = Artifact-released; CC = Contamination-checked; JP = Judge-version-pinned; NR = Not-reported; combined codes (e.g., AR+JP, CC+AR) indicate multiple properties satisfied.

Table B.1. Representative per-paper mapping to all seven taxonomy dimensions (D1–D7). Ten papers from the 151-paper corpus are shown, one per major category, selected to span the three archetypes (Section 8.10 of the main paper) and the five paper categories of Table 8 of the main paper. The complete 151-paper machine-readable dataset is included as a review-time snapshot with the submission; a citable tagged version will be released at acceptance to the GitHub URL above (see Data Availability statement in the main paper). D6 and D7 codings reflect the dual-coded inter-rater agreement of Section 7.5 of the main paper (κ = 0.76 and 0.78 respectively).

Paper (year)	D1 Method	D2 Task	D3 Metric	D4 Auto	D5 Scope	D6 Robust.	D7 Reprod.
<i>Prompting Technique Papers</i>							
Wei et al. [132] (2022) Chain- of-Thought	BM	Reasoning	TS	Auto	CM	Sng	AR
Kojima et al. [54] (2022) Zero-Shot CoT	BM	Reasoning	TS	Auto	CM	MAG	AR
Ouyang et al. [85] (2022) In-structGPT/RLHF	HE	Multi-task	Hyb	HiL	SM	Sng	NR
<i>Automated Prompt Optimization Papers</i>							
Zhou et al. [155] (2022) APE	BM	Multi-task	TS	Auto	CM	Sng	AR
<i>LLM-as-Judge / Metric Papers</i>							
Zheng et al. [151] (2023) MT-Bench	LJ	Multi-task	LJ	Auto	CM	MAG	JP+AR
Liu et al. [72] (2023) G-Eval	LJ	Generation	LJ	Hyb	CM	Sng	JP+AR
<i>Benchmark / Dynamic Benchmark Papers</i>							
Lin et al. [65] (2024) Wild-Bench	LJ	Multi-task	LJ	Auto	CM	MAG	CC+AR
Wang et al. [125] (2018) GLUE	BM	Multi-task	TS	Auto	CDs	Sng	AR
<i>Robustness / Sensitivity Papers</i>							
Zhu et al. [157] (2023) PromptBench	BM	Multi-task	TS	Auto	CM	SB	AR

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Table B.1 (continued)

Paper (year)	D1 Method	D2 Task	D3 Metric	D4 Auto	D5 Scope	D6 Robust.	D7 Reprod.
LLM Evaluation Survey Papers							
Chang et al. [15] (2023) LLM HE Eval Survey		Multi- task	Hyb	HiL	CD	NR	NR

C Full Per-Paper Mapping (151 Papers, All Seven Dimensions)

For full transparency, this appendix presents the complete corpus of 151 papers coded across all seven taxonomy dimensions (D1–D7). The representative ten-paper sample in Appendix B above (Table B.1) provides category-grouped expanded notes; the underlying machine-readable spreadsheet is included as a review-time snapshot with the submission; a citable tagged version will be released at acceptance to <https://github.com/rohithreddybc/PromptEvalTaxonomy> (see Data Availability statement in the main paper). Papers 79–98 were added in the first expansion round, 99–109 in the second, 110–144 in the third, and 145–151 in the fourth (Section 7 of the main paper, Table 4 therein), targeting LLM-judge quality, benchmark contamination, multi-prompt sensitivity, agentic, multilingual, clinical, prompt-injection, contamination-resistant, and CoT-faithfulness work underrepresented in the primary search results. D6 and D7 entries reflect the dual-coded inter-rater agreement of Section 7.5 of the main paper ($\kappa = 0.76$ and 0.78 respectively). Abbreviations follow those used in Appendix B above: D6 = Sng (Single-prompt), MAg (Multi-prompt aggregate), SB (Sensitivity-bounded), NR (Not reported); D7 = AR (Artifact-released), CC (Contamination-checked), JP (Judge-version-pinned), NR (Not reported); combined codes such as “CC+AR” or “JP+AR” indicate that both properties are satisfied.

Table C.1. Comprehensive per-paper mapping of the 151-paper corpus across all seven taxonomy dimensions. D1 (Evaluation Method), D2 (Task Scope), D3 (Metric Type), D4 (Automation Level), D5 (Evaluation Scope), D6 (Robustness), D7 (Reproducibility). D6 and D7 inter-rater $\kappa = 0.76$ and 0.78 respectively. Papers 1–78 were retained from the primary search round; papers 79–98 added in expansion round 1 (LLM-judge quality, contamination, multi-prompt sensitivity); 99–109 in round 2 (agentic, multilingual, prompt-injection, CoT-faithfulness); 110–144 in round 3 (foundation-model reports, RAG, safety, fairness, calibration, clinical, legal); and 145–151 in round 4 (long-context, tool-use, reward-model evaluation). See Table 4 of the main paper for the full expansion history.

#	Paper	System / Topic	D1 Method	D2 Task	D3 Metric	D4 Auto.	D5 Scope	D6 Rob.	D7 Repro.
<i>Core Prompting Technique Papers</i>									
1	Wei et al. [132]	Chain-of-Thought	Benchmark	Reasoning	Task-specific	Automated	Cross-model	Sng	AR
2	Kojima et al. [54]	Zero-Shot CoT	Benchmark	Reasoning	Task-specific	Automated	Cross-model	MAg	AR
3	Wang et al. [128]	Self-Consistency	Benchmark	Reasoning	Task-specific	Automated	Cross-model	Sng	AR
4	Yao et al. [140]	Tree of Thoughts	Benchmark	Reasoning	Task-specific	Automated	Single-model	Sng	AR
5	Zhou et al. [153]	Least-to-Most	Benchmark	Reasoning	Task-specific	Automated	Cross-model	Sng	AR
6	Besta et al. [9]	Graph of Thoughts	Benchmark	Reasoning	Task-specific	Automated	Single-model	Sng	AR
7	Wang et al. [126]	Plan-and-Solve	Benchmark	Reasoning	Task-specific	Automated	Single-model	Sng	AR
8	Lyu et al. [74]	Faithful CoT	Benchmark	Reasoning	Task-specific	Automated	Cross-model	Sng	AR
9	Besta et al. [10]	Chains/Trees/Graphs	Benchmark	Reasoning	Task-specific	Automated	Cross-model	NR	NR
10	Lewis et al. [59]	RAG	Benchmark	QA	Task-specific	Automated	Single-model	Sng	AR
11	Shin et al. [106]	AutoPrompt	Benchmark	QA, Reasoning	Task-specific	Automated	Single-model	Sng	AR
12	Agarwal et al. [3]	Many-Shot ICL	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
13	Ouyang et al. [85]	InstructGPT / RLHF	Human eval	Multi-task	Hybrid	Human-in-loop	Single-model	Sng	NR
14	Wei et al. [22]	FLAN	Benchmark	Multi-task	Task-specific	Automated	Single-model	Sng	AR
15	Li et al. [61]	Structured CoT	Benchmark	Code	Task-specific	Automated	Cross-model	Sng	AR
16	Ferrão et al. [32]	Manual Prompt Study	Benchmark	Code	Task-specific	Hybrid	Single-model	Sng	AR
17	Agrawal et al. [4]	GEPA	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
18	Agarwal et al. [2]	PromptWizard	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
19	White et al. [133]	Prompt Pattern Catalog	Ablation	Multi-task	Task-specific	Manual	Single-model	NR	NR
20	Dong et al. [26]	In-Context Learning	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
<i>Automated Prompt Optimization Papers</i>									
21	Zhou et al. [155]	APE	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR

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Table C.1 (continued)

#	Paper	System / Topic	D1 Method	D2 Task	D3 Metric	D4 Auto.	D5 Scope	D6 Rob.	D7 Repro.
22	Pryzant et al. [87]	APO	Benchmark	Multi-task	Task-specific	Automated	Single-model	Sng	AR
23	Deng et al. [25]	RLPrompt	Benchmark	Multi-task	Task-specific	Automated	Single-model	Sng	AR
24	Li et al. [62]	Auto Prompt Survey	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
25	Ramnath et al. [92]	Auto Prompt Opt. Survey	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
26	He et al. [42]	Context Engineering	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
27	Schulhoff et al. [102]	The Prompt Report	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
<i>Prompt Engineering Survey Papers</i>									
28	Sahoo et al. [97]	PE Techniques Survey	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
29	Vatsal & Dubey [121]	PE Methods Survey	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
30	Liu et al. [73]	PE Taxonomy	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
31	Sasaki et al. [101]	PE Patterns (SE)	Benchmark	Code	Task-specific	Automated	Cross-domain	NR	NR
32	Tian et al. [116]	Prompt Defect Taxonomy	Ablation	Code	Task-specific	Manual	Single-model	NR	NR
33	Chang et al. [15]	LLM Evaluation Survey	Human eval	Multi-task	Hybrid	Human-in-loop	Cross-domain	NR	NR
<i>LLM Evaluation Survey Papers</i>									
34	Guo et al. [41]	LLM Eval. Comprehensive	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
35	Ni et al. [83]	LLM Benchmark Survey	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
36	Mohammadi et al. [82]	LLM Agent Eval. Survey	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
37	Polo et al. [156]	Benchmark ²	Benchmark	Multi-task	Task-specific	Automated	Cross-dataset	MAg	AR
38	Zhao et al. [149]	LLM Explainability	Human eval	Multi-task	Hybrid	Human-in-loop	Cross-domain	NR	NR
39	Wang et al. [125]	GLUE	Benchmark	Multi-task	Task-specific	Automated	Cross-dataset	Sng	AR
<i>Benchmark Papers</i>									

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Table C.1 (continued)

#	Paper	System / Topic	D1 Method	D2 Task	D3 Metric	D4 Auto.	D5 Scope	D6 Rob.	D7 Repro.
40	Wang et al. [124]	SuperGLUE	Benchmark	Multi-task	Task-specific	Automated	Cross-dataset	Sng	AR
41	Srivastava et al. [109]	BIG-Bench	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAG	AR
42	Hendrycks et al. [43]	MMLU	Benchmark	Multi-task	Task-specific	Automated	Cross-dataset	Sng	AR
43	Suzgun et al. [113]	BIG-Bench Hard	Benchmark	Reasoning	Task-specific	Automated	Cross-dataset	Sng	AR
44	Zhu et al. [157]	PromptBench	Benchmark	Multi-task	Task-specific	Automated	Cross-model	SB	AR
45	Wang et al. [129]	MMLU-Pro	Benchmark	Multi-task	Task-specific	Automated	Cross-dataset	Sng	AR
46	Lin et al. [65]	WildBench	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	MAG	CC+AR
47	Gao et al. [36]	RAG Survey	Benchmark	QA	Task-specific	Automated	Cross-domain	NR	NR
48	Zheng et al. [151]	MT-Bench / Chatbot Arena	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	MAG	JP+AR
<i>LLM-as-Judge Papers</i>									
49	Dubois et al. [27]	AlpacaEval 2.0	LLM-as-Judge	Generation	LLM-judge	Automated	Cross-model	Sng	JP+AR
50	Liu et al. [72]	G-Eval	LLM-as-Judge	Generation	LLM-judge	Hybrid	Cross-model	Sng	JP+AR
51	Gu et al. [39]	LLM-as-Judge Survey	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-domain	NR	NR
52	Li et al. [60]	LLM Judge Survey	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-domain	NR	NR
53	Feng et al. [31]	SAGE	LLM-as-Judge	Multi-task	LLM-judge	Hybrid	Cross-model	MAG	JP+AR
54	Tan et al. [114]	JudgeBench	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	MAG	JP+AR
55	Bavaresco et al. [7]	LLMs vs. Humans Study	LLM-as-Judge, Human	Multi-task	LLM-judge	Hybrid	Cross-model	Sng	AR
56	Calderon et al. [13]	Alternative Annotator	LLM-as-Judge, Human	Multi-task	LLM-judge	Hybrid	Cross-model	MAG	AR
57	Chen et al. [17]	MAJ-EVAL	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-domain	MAG	AR
58	Yang et al. [137]	LLM Judge Empirical	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	MAG	AR
59	Ye et al. [141]	Judgement Bias Study	LLM-as-Judge, Human	Multi-task	LLM-judge	Hybrid	Cross-model	Sng	AR
60	Thakur et al. [115]	Judging the Judges	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	Sng	AR
61	Chiang et al. [21]	Can LLMs Replace Humans	LLM-as-Judge, Human	Generation	LLM-judge	Hybrid	Single-model	Sng	AR
62	Min et al. [79]	FActScore	LLM-as-Judge	Generation	LLM-judge	Hybrid	Cross-model	Sng	AR
<i>Metric and Evaluation Quality Papers</i>									
63	Xu et al. [136]	InstructScore	LLM-as-Judge	Generation	LLM-judge	Automated	Cross-model	Sng	AR

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Table C.1 (continued)

#	Paper	System / Topic	D1 Method	D2 Task	D3 Metric	D4 Auto.	D5 Scope	D6 Rob.	D7 Repro.
64	Es et al. [28]	RAGAs	Benchmark	QA	Task-specific	Automated	Single-model	Sng	AR
65	Gehrmann et al. [37]	NLG Eval. Survey	Human eval	Generation	Model-based	Human-in-loop	Cross-domain	NR	NR
66	Zhong et al. [152]	Multi-Dim. Evaluator	Human eval	Generation	Hybrid	Hybrid	Cross-model	Sng	AR
67	Yu et al. [145]	DeCE	LLM-as-Judge	QA	LLM-judge	Automated	Cross-model	Sng	AR
68	Farquhar et al. [56]	Semantic Entropy	Benchmark	QA	Task-specific	Automated	Cross-model	Sng	AR
69	Gao et al. [89]	NLG Eval Status	LLM-as-Judge	Generation	LLM-judge	Automated	Cross-domain	NR	NR
70	Ram et al. [91]	In-Context RAG	Benchmark	QA	Task-specific	Automated	Cross-model	Sng	AR
71	Sciar et al. [103]	Prompt Format Sensitivity	Benchmark	Multi-task	Task-specific	Automated	Cross-model	SB	AR
<i>Prompt Sensitivity and Format Papers</i>									
72	Razavi et al. [93]	PromptSET	Benchmark	QA	Task-specific	Automated	Cross-model	SB	AR
73	Salinas & Gama [100]	Prompt Formatting Study	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAG	AR
74	Salinas & Morstatter [99]	Butterfly Effect Study	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAG	AR
75	Chen et al. [18]	Prompt Alchemy	Benchmark	Code	Task-specific	Automated	Cross-model	Sng	AR
76	He et al. [118]	Secure Code Prompting	Benchmark	Code	Task-specific	Hybrid	Cross-model	Sng	AR
<i>Code and Software Engineering Papers</i>									
77	Hou et al. [44]	LLMs for SE Survey	Benchmark	Code	Task-specific	Automated	Cross-model	NR	NR
78	Khojah et al. [51]	CodePromptEval	Benchmark	Code	Task-specific	Hybrid	Single-model	Sng	AR
<i>LLM-as-Judge Bias and Reliability Studies</i>									
79	Koo et al. [55]	Cognitive Biases in LLM Judges	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	MAG	AR
80	Ye et al. [142]	Justice or Prejudice	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	MAG	AR
81	Zhu et al. [158]	JudgeLM	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	MAG	AR
82	Verga et al. [123]	Panel-of-Judges Evaluation	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	MAG	AR
83	Wataoka et al. [130]	Self-Preference Bias in LLM-as-Judge	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	Sng	NR
<i>Benchmark Contamination and Dynamic Benchmarking Papers</i>									
84	Sainz et al. [98]	NLP Evaluation in Trouble	Benchmark	Multi-task	Task-specific	Automated	Cross-dataset	NR	CC
85	Deng et al. [24]	Investigating Data Contamination	Benchmark	Multi-task	Task-specific	Automated	Cross-dataset	NR	CC

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Table C.1 (continued)

#	Paper	System / Topic	D1 Method	D2 Task	D3 Metric	D4 Auto.	D5 Scope	D6 Rob.	D7 Repro.
86	Yang et al. [138]	Rethinking Benchmark Contamination	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAG	CC
87	Wang et al. [127]	Benchmark Self-Evolving	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	CC+AR
88	McIntosh et al. [78]	LLM Benchmark Inadequacies	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
89	Zeng et al. [148]	LLMBar (Eval. Instruction Following)	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	Sng	AR
90	Chern et al. [20]	ScaleEval (Agent Debate)	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	MAG	AR
91	Li et al. [63]	LLM-based NLG Evaluation Survey	Benchmark	Generation	Model-based	Automated	Cross-model	NR	NR
92	Zhou et al. [154]	Instruction-Following Eval (IFEval)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
93	Qin et al. [88]	ChatGPT Sentiment Reasoning	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
Multi-Prompt, Clinical, and Multimodal Evaluation Papers									
94	Mizrahi et al. [81]	State of What Art (TACL)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAG	AR
95	Sivarajkumar et al. [107]	Clinical Prompting Evaluation (JMIR)	Benchmark	Clinical	Task-specific	Automated	Cross-model	MAG	NR
96	Polo et al. [86]	PromptEval (Multi-Prompt Eval)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	SB	AR
97	Jang & Lukasiewicz [49]	Consistency Analysis of ChatGPT	Human eval	Multi-task	Hybrid	Manual	Cross-domain	NR	NR
98	Fu et al. [33]	MME Multimodal Evaluation	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
Judge Sensitivity and Multilingual Evaluation Papers									
99	Belliblatu et al. [8]	JudgeSense (Judge Sensitivity)	Benchmark	Multi-task	LLM-judge	Automated	Cross-model	SB	AR
100	Vatsal & Singh [122]	Multilingual PE Survey	Benchmark	Multilingual	Task-specific	Automated	Cross-domain	NR	NR
Agentic, Tool-Use, and Safety Evaluation Papers									
101	Liu et al. [70]	AgentBench	Benchmark	Agentic	Task-specific	Automated	Cross-model	Sng	AR
102	Ma et al. [75]	AgentBoard	Benchmark	Agentic	Task-specific	Automated	Cross-model	MAG	AR
103	Liu et al. [71]	HouYi (Prompt Injection)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAG	AR
104	Maloyan et al. [77]	Judge Vulnerability to Injection	LLM-as-Judge	Multi-task	LLM-judge	Automated	Cross-model	MAG	AR
Chain-of-Thought Faithfulness and Hallucination Studies									

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Table C.1 (continued)

#	Paper	System / Topic	D1 Method	D2 Task	D3 Metric	D4 Auto.	D5 Scope	D6 Rob.	D7 Repro.
105	Lanham et al. [58]	CoT Faithfulness Measurement	Benchmark	Reasoning	Task-specific	Automated	Cross-model	MAG	AR
106	Chen et al. [19]	Reasoning Models Faithfulness	Benchmark	Reasoning	Task-specific	Automated	Cross-model	SB	AR
107	Akbar et al. [5]	HalluMeasure	LLM-as-Judge	Generation	LLM-judge	Automated	Cross-model	Sng	AR
108	Liu et al. [67]	RH-Bench (Multimodal Hallu.)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAG	AR
<i>Multilingual LLM-as-Judge Meta-Evaluation</i>									
109	Son et al. [108]	MM-Eval (Multilingual Judge)	LLM-as-Judge	Multilingual	LLM-judge	Automated	Cross-model	MAG	AR
<i>Foundation-Model Technical Reports</i>									
110	Achiam et al. [1]	GPT-4 Technical Report	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	Sng	NR
111	Touvron et al. [119]	LLaMA Foundation Models	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
112	Touvron et al. [120]	Llama 2 Open Foundation Models	Benchmark	Multi-task	Hybrid	Hybrid	Cross-model	MAG	AR
113	Hurst et al. [47]	GPT-4o System Card	Benchmark	Multi-task	Hybrid	Automated	Cross-domain	Sng	NR
114	Bubeck et al. [12]	Sparks of AGI (GPT-4 study)	Human eval	Multi-task	Hybrid	Manual	Cross-domain	Sng	NR
<i>Mathematical Reasoning Benchmarks</i>									
115	Cobbe et al. [23]	GSM8K Grade-School Math	Benchmark	Reasoning	Task-specific	Automated	Cross-model	Sng	AR
116	Lin et al. [66]	TruthfulQA	Benchmark	Multi-task	Task-specific	Hybrid	Cross-model	Sng	AR
117	Mirzadeh et al. [80]	GSM-Symbolic	Benchmark	Reasoning	Task-specific	Automated	Cross-model	SB	AR
118	Shi et al. [105]	MGSM (Multilingual CoT)	Benchmark	Multilingual	Task-specific	Automated	Cross-model	Sng	AR
<i>Code Generation Evaluation Papers</i>									
119	Jimenez et al. [50]	SWE-bench	Benchmark	Code	Task-specific	Automated	Cross-model	Sng	AR
120	Liu et al. [68]	EvalPlus / HumanEval+	Benchmark	Code	Task-specific	Automated	Cross-model	MAG	AR
121	Jain et al. [48]	LiveCodeBench	Benchmark	Code	Task-specific	Automated	Cross-model	Sng	CC+AR
122	Rozière et al. [95]	Code Llama	Benchmark	Code	Task-specific	Automated	Cross-model	Sng	AR
<i>Retrieval-Augmented Generation Evaluation Papers</i>									
123	Saad-Falcon et al. [96]	ARES	LLM-as-Judge	QA	LLM-judge	Automated	Cross-model	MAG	AR

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Table C.1 (continued)

#	Paper	System / Topic	D1 Method	D2 Task	D3 Metric	D4 Auto.	D5 Scope	D6 Rob.	D7 Repro.
124	Yu et al. [146]	Auepora RAG Survey	Benchmark	Multi-task	Hybrid	Automated	Cross-domain	NR	NR
125	Chen et al. [16]	RGB (RAG Benchmark)	Benchmark	QA	Task-specific	Automated	Cross-model	MAg	AR
126	Xiong et al. [134]	MIRAGE / MedRAG (Medical)	Benchmark	Clinical	Task-specific	Automated	Cross-model	MAg	AR
127	Zhao et al. [150]	RAG-for-AIGC Survey	Benchmark	Multi-task	Hybrid	Automated	Cross-domain	NR	NR
<i>Safety, Red-Teaming, and Jailbreak Evaluation Papers</i>									
128	Wei et al. [131]	Jailbroken (Safety Failure Modes)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAg	AR
129	Yu et al. [147]	GPTFuzzer	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAg	AR
130	Bhardwaj & Poria [11]	RED-EVAL (Chain of Utterances)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAg	AR
131	Huang et al. [46]	Catastrophic Jailbreak	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAg	AR
132	Sharma et al. [104]	Constitutional Classifiers	Benchmark	Multi-task	Task-specific	Automated	Cross-model	SB	AR
133	Yi et al. [144]	Jailbreak Attacks & Defenses Survey	Benchmark	Multi-task	Task-specific	Automated	Cross-domain	NR	NR
<i>Bias and Fairness Evaluation Papers</i>									
134	Gallegos et al. [34]	Bias & Fairness in LLMs Survey	Benchmark	Multi-task	Hybrid	Hybrid	Cross-domain	NR	NR
135	Esiobu et al. [29]	ROBBIE (Bias Evaluation)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAg	AR
<i>Calibration and Uncertainty Quantification Papers</i>									
136	Xiong et al. [135]	LLM Confidence Elicitation	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAg	AR
137	Geng et al. [38]	Confidence Calibration Survey	Benchmark	Multi-task	Hybrid	Automated	Cross-domain	NR	NR
138	Steyvers et al. [110]	Calibration Gap (Nature MI)	Human eval	Multi-task	Hybrid	Hybrid	Cross-model	MAg	NR
<i>Legal Domain Evaluation Papers</i>									
139	Guha et al. [40]	LegalBench	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
140	Fei et al. [30]	LawBench (Chinese Legal)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
<i>Medical and Cognitive Evaluation Papers</i>									
141	Nori et al. [84]	GPT-4 on Medical Challenges	Benchmark	Clinical	Task-specific	Hybrid	Single-model	MAg	CC

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Table C.1 (continued)

#	Paper	System / Topic	D1 Method	D2 Task	D3 Metric	D4 Auto.	D5 Scope	D6 Rob.	D7 Repro.
142	Strachan et al. [111]	Theory of Mind (Nature HB)	Human eval	Multi-task	Hybrid	Hybrid	Cross-model	MAG	AR
143	Chang et al. [14]	Red Teaming ChatGPT in Medicine	Human eval	Clinical	Hybrid	Manual	Cross-model	MAG	NR
144	Kim et al. [53]	Medical Hallucinations Eval	Benchmark	Clinical	Task-specific	Hybrid	Cross-model	MAG	AR
Long-Context Evaluation Papers									
145	Hsieh et al. [45]	RULER (Real Context Size)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	SB	AR
146	Bai et al. [6]	LongBench (Bilingual)	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
147	Yen et al. [143]	HELMET (Long-Context Framework)	Benchmark	Multi-task	Hybrid	Hybrid	Cross-model	MAG	AR
Tool-Use and Function-Calling Evaluation Papers									
148	Yao et al. [139]	τ -bench (Tool-Agent-User)	Benchmark	Agentic	Task-specific	Automated	Cross-model	MAG	AR
149	Liu et al. [69]	ToolACE (Function Calling)	Benchmark	Agentic	Task-specific	Automated	Cross-model	MAG	AR
Reward Model and Preference Evaluation Papers									
150	Lambert et al. [57]	RewardBench	Benchmark	Multi-task	Task-specific	Automated	Cross-model	Sng	AR
151	Malik et al. [76]	RewardBench 2	Benchmark	Multi-task	Task-specific	Automated	Cross-model	MAG	AR

D Evaluation Tools and Resources (full mapping)

Section 6 of the main paper summarizes the twelve widely used open evaluation frameworks. The full mapping (tool → supported D-dims, archetype, primary limitation) is given in Table D.2. Archetype legend: A = Benchmark-Automation, B = Judge-Mediated, C = Expert-Anchored, R = Robustness tools (cross-archetype).

Table D.2. Key evaluation tools and their dimensional coverage. No single tool covers all seven dimensions; D6 and D7 are most commonly unaddressed.

Tool	D-dims supported	Arch.	Primary limitation
lm-eval-harness [35]	D1=Benchmark, D4=Automated, D5=Cross-model	A	Closed-form tasks only; narrow construct validity for open-ended generation
Stanford HELM [64]	D1=Benchmark, D4=Automated, D5=Cross-model/dataset, partial D6	A	Computationally expensive; corpus papers often use only subsets
OpenCompass (2023)	D1=Benchmark, D2=Multilingual, D5=Cross-model	A	Partial multilingual coverage; community-maintained model roster
MT-Bench / FastChat [151]	D1=LLM-judge, D3=LLM-judge, D7=JP	B	GPT-4 dependency; verbosity bias in pairwise scoring
AlpacaEval 2.0 [27]	D1=LLM-judge, D3=LLM-judge, D7=AR	B	Length-controlled win-rate still biased toward longer outputs on some domains
Prometheus / Prometheus-2 [52]	D1=LLM-judge, D3=LLM-judge, D7=JP	B	Fine-tuned on GPT-4 feedback; may inherit GPT-4 biases
WildBench [65]	D1=LLM-judge, D7=CC	B	Live leaderboard; specific snapshot reproducibility requires archiving
MMLU-Pro [129]	D1=Benchmark, D3=Task-specific	A	Not dynamically contamination-resistant; harder distractors do not eliminate surface-pattern solving
PromptBench [157]	D6=Sensitivity-bounded (task-solver)	R	Perturbation families predefined; novel adversarial types require extension
PromptSET [93]	D6=Sensitivity-bounded (task-solver)	R	Paraphrase variance only; does not address judge-side sensitivity
JudgeSense [8]	D6=Sensitivity-bounded (judge-side)	R	2026 arXiv preprint; four judge tasks; JSS not yet a community standard
PromptEval [86]	D6=Multi-prompt aggregate	R	Estimates full distribution but does not pinpoint which perturbation families drive variance

These tools partition the design space along the two tensions: benchmark frameworks (Archetype A) maximize reliability with narrow construct validity; LLM-as-Judge frameworks (Archetype B) broaden coverage but introduce judge-side variance that robustness tools begin to quantify.

E Detailed Limitations of the Survey

The five typical limitations summarized in Section 9.5 of the main paper are detailed below.

E.1 Search-String Coverage Gaps

Search strings operate on controlled vocabulary in title, abstract, and keyword fields; papers using alternative or emerging terminology may have been missed. Citation searching recovered 22 additional papers in the primary round; four targeted expansion rounds (Table 5 of the main paper) recovered 73 more (20+11+35+7), but completeness cannot be guaranteed. Because expansion rounds used narrow topical queries rather than full sweeps, they introduce selection bias toward targeted areas; distributional findings are therefore descriptive of *this corpus under these inclusion*

and expansion criteria, not field-level estimates. Scarcities (notably clinical and multilingual primary targets) are properties of these criteria, not claims about the field at large.

E.2 Recency Bias Against 2025–2026 Work

Papers from 2025–2026 without citations or a recognized venue fall outside retention criteria; the corpus may underrepresent recent methodological developments in LLM-as-Judge and dynamic-benchmark subfields. The expansion rounds extended search cutoffs to mitigate this, but the citation-threshold recency bias remains. Two 2026 arXiv preprints [8, 156] are treated as preliminary and never used as sole support for a claim.

E.3 Coding-Judgment Ambiguity at Boundary Cases

D4, D6, and D7 required interpretation when papers did not explicitly describe these properties. The inter-rater statistics in Section 7.5 of the main paper quantify the resulting uncertainty per dimension; the sensitivity analysis in Section 7.6 shows how much propagates to headline claims.

E.4 Corpus Scope Does Not Generalize to All PE Evaluation

The findings apply only to papers matching the inclusion criteria described in Section 7 of the main paper. In particular, the corpus excludes domain-specific applied papers (e.g., clinical case studies that do not generalize beyond a single narrow dataset) and non-English papers. Patterns that appear structural in the corpus may not hold in the excluded literature, and absence-of-evidence findings (such as the low reporting rate of clinical or multilingual PE evaluation) should not be read as evidence-of-absence at the field level.

E.5 Expansion-Round Selection Bias

The four expansion rounds used narrower topical queries on underrepresented areas (LLM-judge quality, agentic evaluation, foundation-model reports, long-context tasks). This introduces directional selection bias: targeted topics are over-represented relative to their true field prevalence, and topics absent from all five rounds remain underrepresented. Distributional findings should not be extrapolated to topic areas absent from the five rounds; cross-round prevalence comparisons should be interpreted with this bias in mind.

E.6 Expansion-Round Audit Table

Table E.3 audits each expansion round individually: what coverage gap triggered it, whether its role was exploratory or confirmatory, how many papers it retained, and the specific directional bias it introduces. Every round over-represents its own target area relative to the field's true prevalence, which is the mechanism behind the selection bias described in E.5.

F Worked Example: Designing a New Clinical RAG Evaluation

Section 5.2 of the main paper summarizes the seven-decision design walkthrough for a new RAG technique on clinical question answering. The fuller per-dimension reasoning is given here.

D1 - Evaluation method. Curated gold-answer sets for clinical QA are scarce, domain-constrained, and legally sensitive; human or hybrid evaluation is more defensible than D1 = Benchmark, which should document why the chosen benchmark covers clinical accuracy.

D2 - Task scope. D2 = Clinical makes the sparsely-represented positioning explicit and signals that general-NLP evaluation instruments may not transfer without adaptation.

D3 - Metric. Reference-based metrics (BLEU, ROUGE) are structurally inadequate without a gold reference; model-based factual precision (e.g., FActScore [79]) is defensible with a well-defined

Table E.3. Expansion-round audit: trigger, role, and bias-risk per supplementary round. All four rounds used the same six databases and inclusion criteria as Round 0.

Rd	Trigger (coverage gap)	Role	Ret.	Bias risk	
1	Few papers studying judge reliability despite 30 corpus papers using LLM-judge methods	Exploratory	20	Over-represents quality / bias	LLM-judge
2	Agentic, multilingual, prompt-injection, CoT-faithfulness gaps	Exploratory	11	Over-represents agentic and multilingual eval	
3	Foundation-model reports, RAG, safety, fairness, calibration, clinical, legal under-covered	Confirmatory + exploratory	35	Over-represents foundation-model and domain-specific	
4	Long-context, tool-use, reward-model gap during 2026-Q1 coding	Exploratory	7	Over-represents benchmarks	2026-recent

knowledge source. D3 = LLM-judge is an alternative but requires calibration against clinician judgments before scores carry clinical interpretability.

D4 - Automation level. Full automation reduces cost but risks systematic errors clinicians would catch. Human-in-loop is more appropriate for a first evaluation where no calibrated automated metric exists.

D5 - Evaluation scope. D5 = Single-model gives limited generalizability evidence; D5 = Cross-model is the stronger basis for deployment claims across systems.

D6 - Robustness reporting. Clinical RAG is exposed to prompt sensitivity because retrieved-context phrasing varies across patient records. D6 = Sensitivity-bounded (variance across paraphrased clinical queries on a held-out set) is defensible; D6 = Single-prompt should be flagged as a generalizability risk.

D7 - Reproducibility commitments. *Judge-version-pinned* (for LLM components) and *contamination-checked* (test set not in training corpus) are stronger choices than D7 = Not-reported, especially since medical benchmarks are smaller and easier to contaminate.

Making these decisions explicit before evaluation reduces post-hoc rationalization and makes the study directly comparable to prior work.

G Future Research Agenda (full mapping)

The seven research directions summarized in Section 9.6 of the main paper map to gaps, dimensions, and supporting references as follows.

H Cluster Validation Details (full)

Section 8.10.1 of the main paper summarizes four cluster-validation checks; this appendix gives the full method and results.

A reviewer could reasonably ask whether the three archetypes reflect structure in the data or labels we imposed. We address this with four complementary checks: a silhouette score against random labellings, an unsupervised *k*-means run with no knowledge of the rule-based labels, stability across alternative clustering methods (hierarchical, GMM, spectral), and a non-parametric bootstrap of cluster membership. Checks 1–3 use each paper’s D1–D7 codes encoded as a one-hot vector with Hamming distance (RNG seed 42 where applicable, matching `recompute_all.py`); the *k*-means silhouette curve reported in Check 2 below uses a separate, narrower one-hot encoding restricted to D2, D4, D5, and D6 with Euclidean distance (seed 42, $n_{\text{init}} = 10$), as documented in `recompute_all.py`’s `appendix_h_curve` function; the two encodings are not the same and should not be conflated.

Table G.4. Seven open research directions derived from the taxonomy tensions. Dims: primary dimensions affected. References are representative corpus instances or foundational work.

Direction	Core gap	Dims	Key refs
Domain-specific frame-works	Expert-calibrated metrics for clinical, legal, multilingual PE	D2,D3	[40, 65, 107]
LLM-judge standard	Calibration disclosure: judge identity, prompt, calibration outcome	D1,D3	[13, 20, 123]
Scope as minimum bar	Multi-model, multi-benchmark, multi-paraphrase by default	D5,D6	[81, 93, 103]
Robustness reporting	Sensitivity bounds alongside point estimates	D6	[8, 86]
Agentic/multimodal conventions	D1/D3/D6 conventions for multi-step and vision-grounded evaluation	D1,D2,D3	[33, 139]
Reporting templates	Venue-endorsed checklist (model, seed, prompt, judge)	all	[37, 90]
Safety & bias integration	Adversarial and demographic probes as first-class dimensions	D3,D6	[34, 131]

Check 1: silhouette against random baselines. The silhouette [94] score for the rule-based three-archetype labelling, computed on a one-hot encoding of D1–D7 (Hamming distance), on the 140 archetype-fitting papers is **0.414**, against a mean of -0.037 for 100 random three-label assignments of the same cardinality, so the partition is far more separated than chance.

Check 2: agreement with unsupervised k -means. k -means at $k = 3$ (k -means++ init, 10 seeds, lowest-SSE, seed 42) on a one-hot encoding of D1–D7 (Hamming distance) with no knowledge of the rule-based labels agrees with the rule-based labels at $\text{ARI} = 0.50$. As a separate silhouette diagnostic, k -means for $k = 2, 3, 4$ on a one-hot encoding restricted to D2, D4, D5, and D6 (Euclidean distance, seed 42, $n_{\text{init}} = 10$) yields silhouette 0.35 at $k = 3$ (0.30 at $k = 2$, 0.36 at $k = 4$); $k = 4$ marginally maximises this restricted-encoding diagnostic, so we take $k = 3$ on parsimony and interpretability, not because it maximises this diagnostic. The partitions agree moderately rather than recovering a different structure.

Caveat (important). All checks encode the same D1–D7 codes used to derive the rule-based labels, and that encoding contains D1, which the assignment rule reads. The recovery is therefore largely the algorithm reading back its own input: clustering on D1 alone reproduces the archetypes at $\text{ARI} \approx 0.97$, at or above what the five algorithms achieve on the full encoding, whereas clustering the five dimensions the rule does not read (D2, D3, D5–D7) recovers them at $\text{ARI} \approx 0.01$ – 0.02 . The honest conclusion is that the archetypes are a compact summary of D1, separable under this coding, not a latent structure the other dimensions independently corroborate. $\text{ARI} = 0.50$ on the full encoding is moderate, not high.

The three archetypes are therefore not exhaustive (7.3% of the corpus carries mixed configurations, plus small populations of ablation-driven and user-study evaluations), but they cover three coherent regions of the validity-reliability space.

Check 3: stability across clustering methods. On the 140-paper archetype-fitting subset we reran the analysis under four alternatives: hierarchical clustering with Ward linkage, hierarchical clustering with average linkage, Gaussian mixture models (GMM) with diagonal covariance, and

spectral clustering (nearest-neighbours affinity). At $k = 3$, the adjusted Rand index against the rule-based labels was 0.50 (k -means), 0.52 (Ward), 0.90 (average linkage), 0.49 (GMM), and 0.50 (spectral); silhouette scores ranged from 0.41 to 0.49. Average-linkage hierarchical clustering produced the closest agreement with the rule-based labels ($\text{ARI} = 0.90$). Internal k -selection indices do not cleanly favour $k = 3$: the restricted-encoding k -means silhouette curve of Check 2 is marginally higher at $k = 4$ (0.36) than $k = 3$ (0.35), and the gap statistic on uniform-random references rises monotonically through $k = 7$ (0.24 at $k = 3$ to 0.56 at $k = 7$), a known artefact on sparse binary data [112, 117]. We take $k = 3$ on parsimony and interpretability, not because an internal index selects it.

Check 4: bootstrap stability of cluster membership. Across 200 non-parametric bootstrap resamples of the 140-paper subset (sample with replacement, refit k -means at $k = 3$ each iteration, seed 42), the mean within-rule-archetype co-cluster probability was 0.73, 0.89, and 0.65 for Archetypes A, B, and C respectively, against a between-archetype mean of 0.09. The 0.67 separation indicates that papers assigned to the same rule-archetype remain co-clustered under bootstrap resampling at roughly eight times the rate of papers assigned to different archetypes.

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